

Malak Serag Ibrahim

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Summary

Computer Engineering student with hands-on experience in software development, data analytics, and system design. Skilled in building web, mobile, and desktop applications and analyzing data to extract insights. Experienced in academic and industry projects and currently exploring software engineering, data analytics, cybersecurity, and DevOps.

Technical Skills

- **Programming Languages:** C, C++, C#, Java, JavaScript, SQL, Python, 8051 Assembly
- **Web & Mobile Development:** Vue.js, Nuxt.js, Node.js, Flutter, HTML, CSS, Tailwind CSS
- **Machine Learning & Analytics:** Scikit-learn, Pandas, Matplotlib, Seaborn, Linear & Logistic Regression, K-Means Clustering, Feature Engineering, Model Evaluation, Data Cleaning & Visualization
- **Security & Networking:** AES-128 Encryption, Cryptography Fundamentals (S-Box, Key Expansion, GF(2⁸) Arithmetic), Network Security Basics, Data Security Principles
- **Tools & Technologies:** Git, Postman, JSON, Tkinter, Arduino IDE, Proteus, OxyPlot, CodingSeb.ExpressionEvaluator, MATLAB, AutoCAD
- **Hardware & Systems:** 8051 Microcontroller, Arduino, MFRC522 RFID, EEPROM, SPI & Serial Communication, LCD/Servo Interfacing, VHDL, ISE Project Navigator, Digital Design (FSMs, Counters, Flip-Flops), Jupyter Notebook

Experience

- **Full-Stack Developer Intern (Web & Mobile)** July 2025 – August 2025
Misr Technology Services (MTS) Cairo, Egypt
 - Developed 5+ full-stack and mobile applications using Vue.js, Node.js, and Flutter
 - Designed RESTful APIs and responsive user interfaces
 - Managed client-server communication using JSON
 - Delivered projects including a news platform, forum system, and blog
- **IT Trainee** August 2025
Petrojet Cairo, Egypt
 - Gained experience in networking fundamentals and system administration
 - Simulated networks using GNS3 and explored enterprise infrastructure
 - Explored SAP systems and IT infrastructure
- **Computer Science OL IGCSE Teaching Assistant** July 2021 – May 2024
Rowad College Cairo, Egypt
 - Assisted students in algorithms, logic gates, and programming concepts
 - Explained complex technical topics in a simplified and structured manner
- **Math OL/AS IGCSE Teaching Assistant** Nov 2023 – Feb 2024
Rowad College Cairo, Egypt
 - Supported students in mathematical problem-solving and exam preparation
 - Strengthened analytical thinking and communication skills through teaching

Projects

- **Machine Learning & Data Analytics Projects** Mar 2026 – June 2026
Python, Pandas, Scikit-learn, Data Visualization
 - Built Linear Regression models on the [Housing](#) datasets and Logistic Regression classifiers for [Heart Disease](#) risk and [MOOCs](#) urgency (with K-Means clustering for student segmentation), evaluated with MAE, MSE, R^2 , accuracy, precision, recall, and F1-score.
 - Performed EDA on real-world datasets ([sports seasons](#), [global happiness](#)) and applied preprocessing such as missing-value handling, feature scaling, outlier treatment, and feature engineering.
 - Communicated insights through visualizations such as heatmaps, scatter plots, histograms, and boxplots.
 - **Learned:** the end-to-end data analytics workflow—data cleaning, EDA, feature engineering, model building, evaluation, and turning results into clear, data-driven insights.
- **AES-128 Encryption Simulator** June 2026
Python, Tkinter GUI
 - [AES Encryption Repository](#)
 - Implemented AES-128 from scratch in Python (no crypto libraries), including SubBytes, ShiftRows, MixColumns over $GF(2^8)$, AddRoundKey, and full key expansion generating all 11 round keys.
 - Built a Tkinter GUI that visualizes the internal 4×4 state matrix after each step and round, plus a key-expansion mode showing RotWord, SubWord, Rcon, and the g-function.
- **RFID Scanning & Access Control System** June 2026
8051 Assembly, Arduino, Proteus
 - [RFID Access Control Repository](#)
 - Designed a two-processor RFID access control system: an Arduino + MFRC522 reads card UIDs over SPI and verifies them, while an AT89C51 (8051) drives the LCD, servo lock, LEDs, and buzzer.
 - Implemented card verification against UIDs stored in EEPROM with auto-enrollment, and granted or denied access via a serial command link between the two controllers.
 - Designed and simulated the full circuit in Proteus before hardware assembly.
- **Cinema Booking System** Dec 2025
Full-Stack Project
 - [Cinema Booking System Repository](#)
 - Developed a full-stack cinema booking system with real-time seat selection, reservation management, and user interaction handling
 - Implemented backend logic and database integration to efficiently manage seat availability, booking transactions, and system updates
 - Designed an intuitive and user-friendly interface to streamline the booking process and enhance overall user experience
- **News Platform (Web + Mobile)** July 2025
Vue.js, Node.js, Flutter
 - [News website Repository](#)
 - Built a full-stack news platform supporting dynamic content rendering, category-based filtering, and real-time updates
 - Designed and integrated RESTful APIs to enable efficient communication between frontend and backend systems
 - Developed responsive and user-friendly interfaces for both web and mobile platforms to enhance accessibility

- **Web Applications Suite (Forum, Blog, To-Do, Reaction Timer)** July 2025
Full-Stack Applications
 - Developed multiple web applications including a forum system, blog platform, to-do manager, and reaction timer game
 - Implemented features such as user interaction, content management, task tracking, and real-time updates across applications
 - Strengthened full-stack development skills by working on diverse application architectures and handling multiple use cases
- **Graphing & Scientific Calculator** April 2025
C#, OxyPlot
 - [Calculator Repository](#)
 - Developed a scientific calculator capable of evaluating complex mathematical expressions accurately
 - Implemented real-time graph plotting functionality using OxyPlot for visualizing mathematical functions
 - Enhanced usability through an interactive interface supporting dynamic input and real-time visualization
- **Student Management System** March 2025
C++, File Handling
 - Built a system to manage student records efficiently including storing, updating, and retrieving data
 - Implemented CRUD operations using file handling to ensure persistent data storage and manipulation
 - Designed structured data organization for efficient access and improved system performance

Education

- **Arab Academy for Science, Technology & Maritime Transport (AAST)** Expected Jan 2028
B.Sc. in Computer Engineering Cairo, Egypt
- **Rowad College** 2023
IGCSE Diploma Cairo, Egypt